

Supply Chain Analytics — Assignment 1

Advanced Business Statistics

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Course: Advanced Business Statistics

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Abstract

This report summarizes analysis of the DataCo supply chain dataset ($\approx 180k$ transactions). Goals: identify revenue/profit drivers, delivery and supplier risks, shipping mode impacts, and build a baseline predictive model for return/cancellation risk. An interactive Streamlit dashboard is included in the repository.

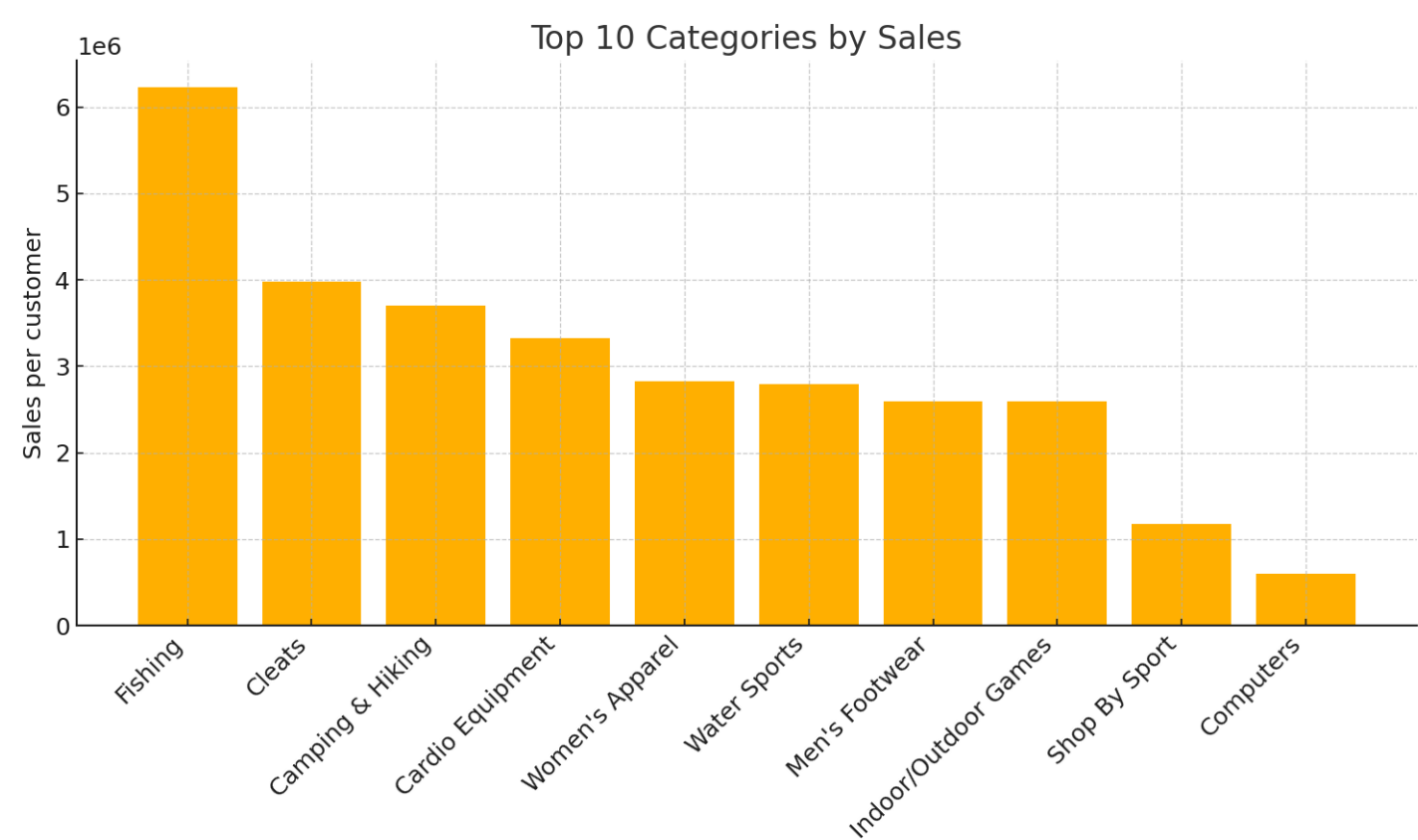
Data & Methods

Data cleaning: missing values handling, date parsing, and feature engineering (benefit-per-order, late flags). Visuals produced using aggregated group-bys and scatter analysis. A baseline logistic regression model was trained for exploratory risk prediction.

Key Findings

1) Revenue and Profit Drivers

Top categories by sales are shown. High-price low-volume items show high benefit per order, while bulk categories drive total sales.



2) Delivery Risk

Shipping days correlate with late delivery risk in the sample. Investigate First Class operational promises.



3) Spatial Insights & Recommendations

A geographic scatter of profit highlights regional clusters. Recommendations: supplier scorecards, review First Class promises, and develop a predictive pipeline.

